

Se-Sp_analysis

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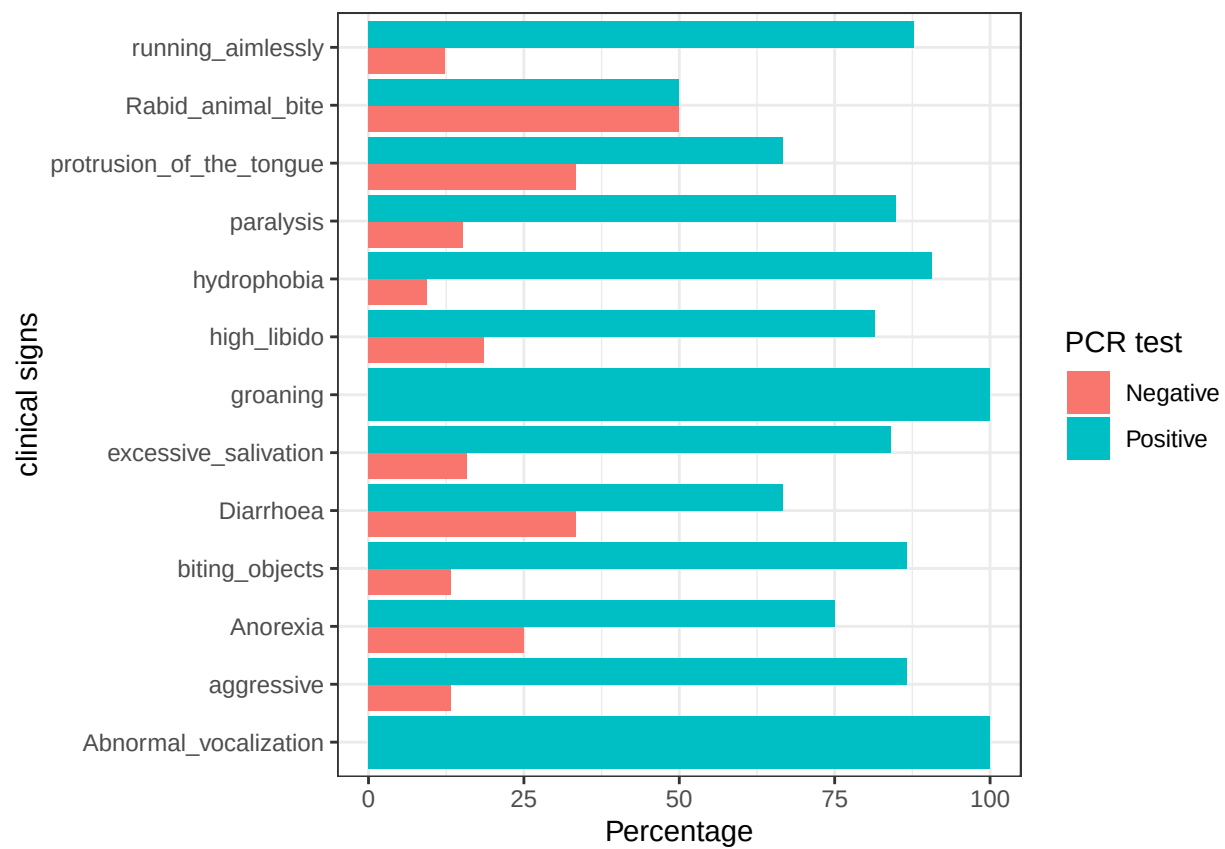
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Se, Sp, NPV, PPV

```
##           Outcome +      Outcome -      Total
## Test +           90           2           92
## Test -            6          14           20
## Total            96          16          112
##
## Point estimates and 95% CIs:
## -----
## Apparent prevalence *           0.82 (0.74, 0.89)
## True prevalence *             0.86 (0.78, 0.92)
## Sensitivity *                 0.94 (0.87, 0.98)
## Specificity *                 0.88 (0.62, 0.98)
## Positive predictive value *    0.98 (0.92, 1.00)
## Negative predictive value *    0.70 (0.46, 0.88)
## Positive likelihood ratio      7.50 (2.05, 27.45)
## Negative likelihood ratio      0.07 (0.03, 0.16)
## False T+ proportion for true D- * 0.12 (0.02, 0.38)
## False T- proportion for true D+ * 0.06 (0.02, 0.13)
## False T+ proportion for T+ *    0.02 (0.00, 0.08)
## False T- proportion for T- *    0.30 (0.12, 0.54)
## Correctly classified proportion * 0.93 (0.86, 0.97)
## -----
## * Exact CIs
```

PCR +VE and -VE of the relevant variables

```
##
## Abnormal_vocalization aggressive Anorexia biting_objects Diarrhoea
## 0           0           10           3           8           1
## 1           4           65           9          52          2
##
## excessive_salivation groaning high_libido hydrophobia paralysis
## 0           10           0           5           3           5
## 1           53           2          22          29          28
##
## protrusion_of_the_tongue Rabid_animal_bite running_aimlessly
## 0           1           1           6
## 1           2           1          43
```



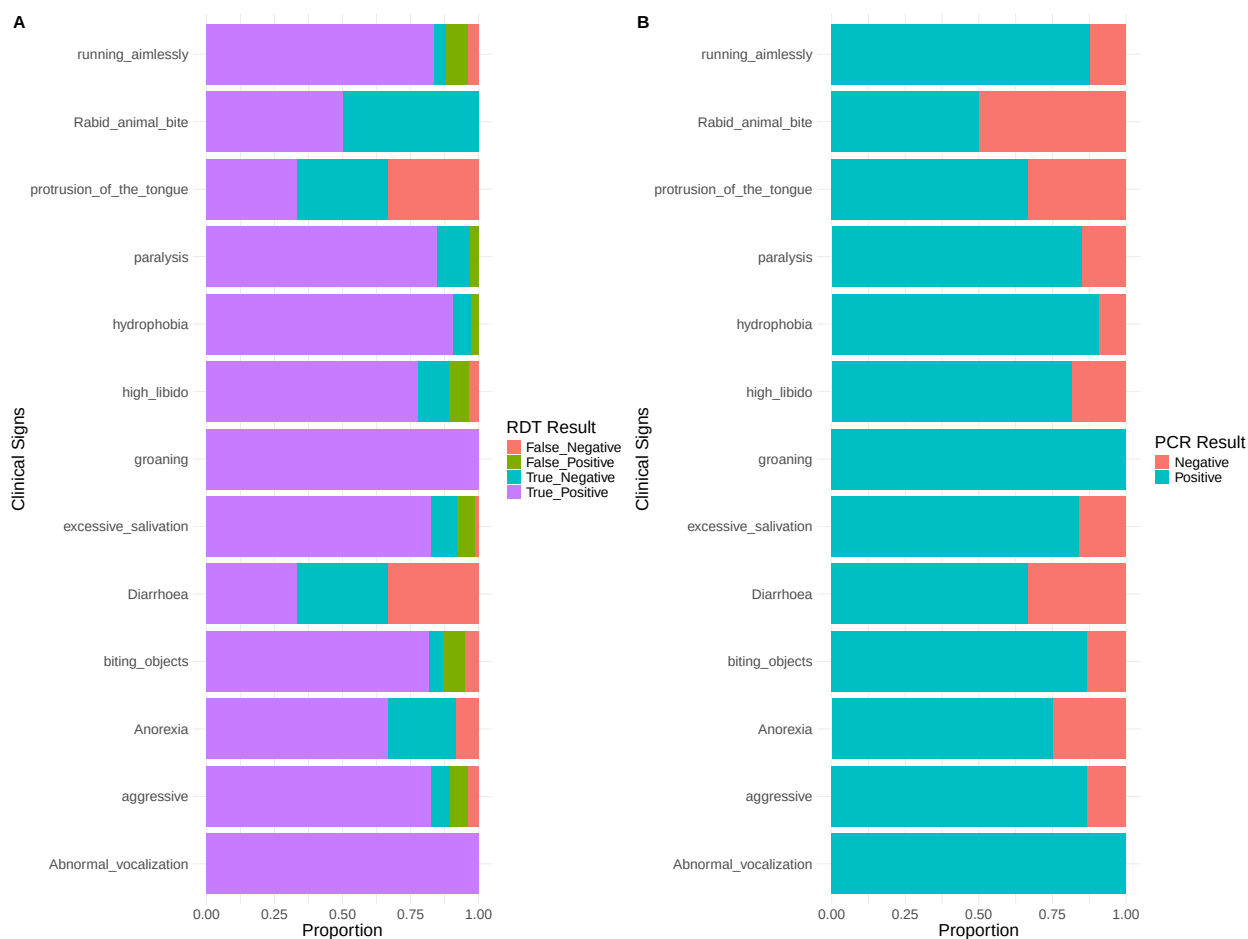


Table species vs the symptoms

species	Anorexia	Diarrhoea	aggressive	biting_objects	excessive_salivation	groaning	high_libido	hydrophobia	running_aimlessly
Dog	3	3	58	48	44	2	18	17	
Goat	6	0	11	11	5	0	5	4	
Cattle	3	0	5	0	13	0	3	10	
Sheep	0	0	1	1	1	0	1	1	
Total	12	3	75	60	63	2	27	32	

GLM

```
##
## Call:
## glm(formula = lab_diagnostics_pcr ~ aggressive, family = "binomial",
##      data = model_glm)
##
## Coefficients:
```

```

##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)  1.5963      0.5118   3.119 0.00181 **
## aggressive   0.1906      0.4801   0.397 0.69141
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 68.275  on 81  degrees of freedom
## Residual deviance: 68.099  on 80  degrees of freedom
## AIC: 72.099
##
## Number of Fisher Scoring iterations: 4

##
## Call:
## glm(formula = lab_diagnostics_pcr ~ biting_objects, family = "binomial",
##      data = model_glm)
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.6722      0.4285   3.903 9.51e-05 ***
## biting_objects  0.1301      0.4337   0.300  0.764
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 68.275  on 81  degrees of freedom
## Residual deviance: 68.177  on 80  degrees of freedom
## AIC: 72.177
##
## Number of Fisher Scoring iterations: 4

##
## Call:
## glm(formula = lab_diagnostics_pcr ~ excessive_salivation, family = binomial(link = "logit"),
##      data = model_glm)
##
## Coefficients:
##           Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.8463      0.4242   4.353 1.35e-05 ***
## excessive_salivation -0.1043      0.3501  -0.298  0.766
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 68.275  on 81  degrees of freedom
## Residual deviance: 68.191  on 80  degrees of freedom
## AIC: 72.191
##
## Number of Fisher Scoring iterations: 4

```

```
##
## Call:
## glm(formula = lab_diagnostics_pcr ~ running_aimlessly, family = binomial(link = "logit"),
##      data = model_glm)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1.5992     0.4119   3.882 0.000104 ***
## running_aimlessly  0.2959     0.5201   0.569 0.569330
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 68.275  on 81  degrees of freedom
## Residual deviance: 67.936  on 80  degrees of freedom
## AIC: 71.936
##
## Number of Fisher Scoring iterations: 4

##
## Call:
## glm(formula = lab_diagnostics_pcr ~ biting_objects + aggressive,
##      family = "binomial", data = model_glm)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1.5943     0.5170   3.084 0.00204 **
## biting_objects  0.0207     0.5988   0.035 0.97242
## aggressive      0.1764     0.6319   0.279 0.78011
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 68.275  on 81  degrees of freedom
## Residual deviance: 68.098  on 79  degrees of freedom
## AIC: 74.098
##
## Number of Fisher Scoring iterations: 4

##
## Call:
## glm(formula = lab_diagnostics_pcr ~ aggressive + biting_objects +
##      excessive_salivation + hydrophobia + running_aimlessly +
##      high_libido + Rabid_animal_bite + paralysis + Anorexia +
##      Diarrhoea + groaning + Abnormal_vocalization, family = binomial(link = "logit"),
##      data = model_glm)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      2.11995     0.94034   2.254 0.0242 *
## aggressive        0.67641     0.86904   0.778 0.4364
## biting_objects    0.10472     0.69549   0.151 0.8803
```

```

## excessive_salivation    -0.82736    0.66479   -1.245    0.2133
## hydrophobia             1.02522    0.75779    1.353    0.1761
## running_aimlessly       0.07981    0.74894    0.107    0.9151
## high_libido             -0.99216    0.75354   -1.317    0.1880
## Rabid_animal_bite       -3.02312    1.84087   -1.642    0.1005
## paralysis               -0.18869    0.77907   -0.242    0.8086
## Anorexia                -0.99365    0.99973   -0.994    0.3203
## Diarrhoea               -1.29409    1.64711   -0.786    0.4321
## groaning                7.91175  1978.09028    0.004    0.9968
## Abnormal_vocalization   15.66569  1971.79996    0.008    0.9937
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 68.275  on 81  degrees of freedom
## Residual deviance: 58.563  on 69  degrees of freedom
## AIC: 84.563
##
## Number of Fisher Scoring iterations: 16

##
## Call:
## glm(formula = lab_diagnostics_pcr ~ aggressive * biting_objects *
##     excessive_salivation, family = binomial(link = "logit"),
##     data = model_glm)
##
## Coefficients:
##
##                Estimate Std. Error z value
## (Intercept)         0.48530    1.04641   0.464
## aggressive           3.28508    1.63832   2.005
## biting_objects      -0.07734    1.59228  -0.049
## excessive_salivation  0.77180    1.09753   0.703
## aggressive:biting_objects -1.52643    0.96011  -1.590
## aggressive:excessive_salivation -2.24131    1.06894  -2.097
## biting_objects:excessive_salivation 1.01704    1.30034   0.782
## aggressive:biting_objects:excessive_salivation 0.37675    0.29447   1.279
##
##                Pr(>|z|)
## (Intercept)         0.6428
## aggressive           0.0449 *
## biting_objects       0.9613
## excessive_salivation 0.4819
## aggressive:biting_objects 0.1119
## aggressive:excessive_salivation 0.0360 *
## biting_objects:excessive_salivation 0.4341
## aggressive:biting_objects:excessive_salivation 0.2008
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##    Null deviance: 68.275  on 81  degrees of freedom
## Residual deviance: 61.807  on 74  degrees of freedom
## AIC: 77.807

```

```
##
## Number of Fisher Scoring iterations: 8

##
## Call:
## glm(formula = lab_diagnostics_pcr ~ aggressive * excessive_salivation,
##      family = binomial(link = "logit"), data = model_glm)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      1.83785    0.84151   2.184   0.029 *
## aggressive        0.21853    0.71909   0.304   0.761
## excessive_salivation -0.38695    0.69668  -0.555   0.579
## aggressive:excessive_salivation  0.04118    0.21367   0.193   0.847
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 68.275  on 81  degrees of freedom
## Residual deviance: 67.696  on 78  degrees of freedom
## AIC: 75.696
##
## Number of Fisher Scoring iterations: 4
```